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Hiring Manager

Benchling

San Francisco, California

Dear Hiring Manager,

I'm Ajay Venkatesh, completing an M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus AI work at NYU's Novel AI Technologies. I'm applying for the Software Engineer, Agents role, where I'd help prototype and productionize end-to-end AI agents that automate scientific workflows inside Benchling's platform.

At Amazon, I built an **AI agent** with **RAG** and an **agentic workflow** that retrieves operational data, reasons over context, and produces actionable output for users. I implemented a **Model Context Protocol (MCP)** server for tool-augmented **LLM** retrieval, provisioned **AWS CDK** infrastructure on **ECS Fargate**, and shipped a **React** and **TypeScript** UI integrated with backend APIs through iterative feedback and evaluations.

At CampusMesh, I productionized **LLM RAG** pipelines for an AI assistant using **embedding-based semantic search** over structured indexes. At NYU, I partnered with research and product stakeholders on full-stack tools in **Python** and **React**, integrated **LangChain** and **GPT-4** document workflows, and refined features based on user feedback and operational data review.

My stack includes **Python**, **TypeScript**, **React**, **Node.js**, **LangChain**, **FAISS**, **PyTorch**, **Hugging Face**, **AWS Bedrock**, **Docker**, and **Git**. Benchling's biotech R&D domain and internal agent platform frameworks are areas I have not shipped in production yet, but my agent, RAG, and full-stack delivery background and willingness to learn from scientists and customers give me a strong foundation to ramp quickly.

What draws me to Benchling is the chance to build AI agents that help researchers move faster from experiment design through analysis and reporting. I care about rapid experimentation, thoughtful evaluation, and collaborating across engineering and science to turn prototypes into dependable product features.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh